

Limited Dependent Variable Models and Sample Selection Corrections

Intermediate Econometrics

Chengye Jia

Department of Economics
Shandong University of Finance and Economics

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Wooldridge, Chapter 17

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Synthesis

Motivation: The Limited Dependent Variable Zoo (§17-0)

Where We Are: From OLS to Nonlinear Models

Through Chapter 16 the dependent variable y was treated as (roughly) **continuous and unrestricted**. OLS and its extensions did the work.

A limited dependent variable (LDV) is a variable whose range is **substantively restricted**:

- ▶ **Binary:** $y \in \{0, 1\}$ — employed or not, approved or not
- ▶ **Fractional:** $y \in [0, 1]$ — the 401(k) participation rate
- ▶ **Count:** $y \in \{0, 1, 2, \dots\}$ — number of arrests in a year
- ▶ **Corner solution:** $y \geq 0$ with a pileup at 0 — hours worked, charitable gifts

Why care? Forcing OLS on these variables can give nonsensical fitted values (negative probabilities, negative hours) and constant partial effects where economics demands diminishing ones.

Two Distinct Problems Live in This Chapter

Problem 1: The variable itself

y is genuinely restricted in the **population** (observed fully):

- ▶ Binary / fractional $\Rightarrow$ logit, probit
- ▶ Count / nonnegative $\Rightarrow$ Poisson
- ▶ Corner solution $\Rightarrow$ Tobit

Problem 2: How we observe y

y would be fine, but the **data** are incomplete:

- ▶ Top coding / censoring $\Rightarrow$ censored reg.
- ▶ Sub-population sampled $\Rightarrow$ truncated reg.
- ▶ Selection on an outcome $\Rightarrow$ Heckit

Keep these separate. Tobit for “hours worked” models economic behavior (people *choose* zero); censored regression for “top-coded wealth” fixes a survey defect. Same algebra — very different interpretation.

Running Application: Married Women's Labor Supply (MROZ)

We thread one dataset through the whole lecture — MROZ, $n = 753$ married women (Mroz, 1987).

One dataset — four LDV faces

- ▶ **Binary:** *inlf* — in the labor force? (428 yes, 325 no) $\Rightarrow$ logit / probit
- ▶ **Corner solution:** *hours* — annual hours worked (325 zeros, positives range 12–4,950) $\Rightarrow$ Tobit
- ▶ **Selection:** $\log(wage)$ — observed *only* for the 428 workers $\Rightarrow$ Heckit
- ▶ **Fractional / count:** mirrored in companion datasets (401(k) *prate*, arrests *narr86*)

Explanatory variables throughout: *nwifeinc* (other income), *educ*, *exper*, *exper*², *age*, *kidslt6* (kids < 6), *kidsge6* (kids 6–18).

Socratic Warm-Up

Q1. A linear probability model predicts $\hat{P}(inlf = 1) = 1.08$ for a highly educated woman. What went wrong, and why can it *never* happen with a probit?

Q2. In the MROZ data, 325 of 753 women report exactly zero hours. Is “hours worked” a censored variable or a corner solution? Does the distinction change the model we fit?

Q3. We only observe wages for women who *work*. If working women are systematically more able, will OLS of $\log(wage)$ on *educ* (using workers only) over- or under-state the return to education?

We will answer all three by the end of the lecture.

Binary Response: Logit and Probit (§17-1)

The Linear Probability Model and Its Discontents

For a binary y , the LPM (Chapter 7) models the **response probability**

$$P(y = 1 \mid \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k.$$

What LPM gets right

- ▶ Coefficients *are* partial effects
- ▶ Consistent for the APE under MLR.1–4
- ▶ Fast; great for treatment effects

What LPM gets wrong

- ▶ Fitted $\hat{P}$ can fall outside $[0, 1]$
- ▶ Partial effect of x_j is *constant*
- ▶ Errors are inherently heteroskedastic

Fix: run the index $\mathbf{x}\beta$ through a function $G(\cdot)$ that is *squashed* into $(0, 1)$. That is exactly what logit and probit do.

Logit and Probit: Squashing the Index into (0, 1)

Binary response model

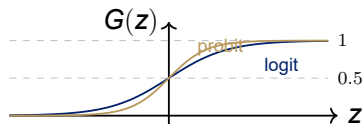
$$P(y = 1 \mid \mathbf{x}) = G(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k) = G(\beta_0 + \mathbf{x}\boldsymbol{\beta}), \quad 0 < G(z) < 1.$$

Logit: logistic CDF

$$\Lambda(z) = \frac{\exp(z)}{1 + \exp(z)}.$$

Probit: normal CDF

$$\Phi(z) = \int_{-\infty}^z \phi(v) dv.$$



Both are increasing CDFs, steepest at $z = 0$. They guarantee $\hat{P} \in (0, 1)$ and *diminishing* partial effects near the extremes.

Where Do Logit and Probit Come From? A Latent Variable

Latent variable model

Let y^* be an unobserved (latent) “propensity”:

$$y^* = \beta_0 + \mathbf{x}\boldsymbol{\beta} + \mathbf{e}, \quad y = \mathbf{1}[y^* > 0].$$

The error $\mathbf{e}$ is **independent** of $\mathbf{x}$ and symmetric about zero.

Then, using symmetry $1 - G(-z) = G(z)$,

$$P(y = 1 \mid \mathbf{x}) = P(y^* > 0 \mid \mathbf{x}) = P(\mathbf{e} > -(\beta_0 + \mathbf{x}\boldsymbol{\beta})) = G(\beta_0 + \mathbf{x}\boldsymbol{\beta}).$$

- ▶ $\mathbf{e} \sim \text{logistic} \Rightarrow \text{logit}$; $\mathbf{e} \sim \text{normal} \Rightarrow \text{probit}$.
- ▶ The latent scale has *no natural units*, so raw β_j are not directly meaningful — only their **sign** and **response-probability** effects.

Partial Effects: Why a Probit Coefficient Is Not a Marginal Effect

For a (roughly) continuous x_j , the chain rule gives

$$\frac{\partial P(y = 1 \mid \mathbf{x})}{\partial x_j} = g(\beta_0 + \mathbf{x}\beta) \beta_j, \quad g(z) \equiv \frac{dG}{dz}(z) > 0.$$

The effect of x_j is β_j **scaled** by the density $g(\beta_0 + \mathbf{x}\beta)$, which *depends on the values of all regressors*. So the partial effect **varies across observations**.

For a **discrete** change in x_1 from 0 to 1:

$$G(\beta_0 + \beta_1 + \beta_2 x_2 + \cdots) - G(\beta_0 + \beta_2 x_2 + \cdots).$$

Anti-pattern: “the probit coefficient *is* the marginal effect.” **Wrong** — it is β_j before scaling by $g(\cdot)$.

Relative effects are clean, though: $\frac{\partial P / \partial x_j}{\partial P / \partial x_h} = \frac{\beta_j}{\beta_h}$ — the scale factor cancels.

Maximum Likelihood Estimation

OLS does not apply — G is nonlinear. We use **MLE**. With $y \mid \mathbf{x}$ Bernoulli, the density for observation i is

$$f(y_i \mid \mathbf{x}_i) = [G(\mathbf{x}_i\boldsymbol{\beta})]^{y_i} [1 - G(\mathbf{x}_i\boldsymbol{\beta})]^{1-y_i}, \quad y_i \in \{0, 1\}.$$

Log-likelihood

$$\ell_i(\boldsymbol{\beta}) = y_i \log[G(\mathbf{x}_i\boldsymbol{\beta})] + (1 - y_i) \log[1 - G(\mathbf{x}_i\boldsymbol{\beta})], \quad \mathcal{L}(\boldsymbol{\beta}) = \sum_{i=1}^n \ell_i(\boldsymbol{\beta}).$$

- ▶ $\hat{\boldsymbol{\beta}}$ maximizes $\mathcal{L}$; no closed form — solved numerically.
- ▶ Under general conditions MLE is **consistent, asymptotically normal, and efficient**.
- ▶ Each $\hat{\beta}_j$ comes with an asymptotic SE $\Rightarrow$ form $t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$ exactly as before.

First-Order Conditions: The Score Equations

$\hat{\beta}$ solves $\partial \mathcal{L} / \partial \beta = \mathbf{0}$. Differentiating $\ell_i = y_i \log G + (1 - y_i) \log(1 - G)$ with $g \equiv G'$:

Score (likelihood) equations

$$\sum_{i=1}^n \frac{[y_i - G(\mathbf{x}_i\beta)] g(\mathbf{x}_i\beta)}{G(\mathbf{x}_i\beta)[1 - G(\mathbf{x}_i\beta)]} \mathbf{x}_i = \mathbf{0}.$$

Logit's clean special case. For $G = \Lambda$ we have $g = \Lambda(1 - \Lambda) = G(1 - G)$, so the weight collapses to 1:

$$\sum_{i=1}^n [y_i - \Lambda(\mathbf{x}_i\beta)] \mathbf{x}_i = \mathbf{0}.$$

The logit score is exactly “residuals orthogonal to the regressors” — the MLE analog of the OLS normal equations. With an intercept in $\mathbf{x}_i$, the fitted probabilities average to the sample frequency: $n^{-1} \sum_i \Lambda(\mathbf{x}_i\beta) = \bar{y}$. No closed form exists; solve numerically (Newton–Raphson).

Testing and Goodness-of-Fit

Likelihood ratio test

For q exclusion restrictions,

$$LR = 2(\mathcal{L}_{ur} - \mathcal{L}_r) \stackrel{a}{\sim} \chi^2_q.$$

The MLE analog of the F test: dropping variables can only lower $\mathcal{L}$.

McFadden's pseudo- R^2

$$\tilde{\rho}^2 = 1 - \frac{\mathcal{L}_{ur}}{\mathcal{L}_0},$$

where $\mathcal{L}_0$ uses an intercept only. Lies in $[0, 1)$ but is **not** a usual R^2 .

Percent correctly predicted: $\tilde{y}_i = 1$ if $G(\hat{\beta}_0 + \mathbf{x}_i\beta) \geq 0.5$. Useful, but *misleading* when one outcome is rare — report the percent correct *within each* outcome ($\hat{q}_0, \hat{q}_1$), not just overall.

Anti-pattern: “pseudo- $R^2 = R^2$.” Not comparable; 0.2–0.4 is already a respectable McFadden value.

MROZ: Labor Force Participation — LPM vs. Logit vs. Probit

Table 17.1 — Dependent variable: *inlf* (in labor force)

Variable	LPM (OLS)	Logit (MLE)	Probit (MLE)
<i>nwifeinc</i>	−0.0034 (0.0015)	−0.021 (0.008)	−0.012 (0.005)
<i>educ</i>	0.038 (0.007)	0.221 (0.043)	0.131 (0.025)
<i>exper</i>	0.039 (0.006)	0.206 (0.032)	0.123 (0.019)
<i>exper</i> ²	−0.00060 (0.00019)	−0.0032 (0.0010)	−0.0019 (0.0006)
<i>age</i>	−0.016 (0.002)	−0.088 (0.015)	−0.053 (0.008)
<i>kidslt6</i>	−0.262 (0.032)	−1.443 (0.204)	−0.868 (0.119)
<i>kidsge6</i>	0.013 (0.014)	0.060 (0.075)	0.036 (0.043)
% correct pred. / pseudo R^2	73.4 / 0.264	73.6 / 0.220	73.4 / 0.221

Signs and significance agree across all three. **Magnitudes do not** — coefficients live on different scales. To compare, we need *partial effects*, not raw coefficients.

Summarizing Effects: APE and PEA (§17-1d)

Two Ways to Get One Number: PEA and APE

The partial effect $g(\beta_0 + \mathbf{x}\beta)\beta_j$ varies by i . We summarize with a single *scale factor*.

Partial Effect at the Average (PEA)

Plug in the **average** regressors, then scale:

$$\text{PEA}_j = g(\hat{\beta}_0 + \bar{\mathbf{x}}\hat{\beta}) \hat{\beta}_j.$$

Problem: “the average person” may not exist (e.g. $\overline{\text{female}} = 0.48$).

Average Partial Effect (APE)

Scale *each* person, then average:

$$\text{APE}_j = \left[\frac{1}{n} \sum_{i=1}^n g(\hat{\beta}_0 + \mathbf{x}_i\hat{\beta}) \right] \hat{\beta}_j.$$

Also called the average marginal effect (AME). **Preferred** in practice.

For **discrete** regressors, do not use the calculus approximation — difference G directly and average: $n^{-1} \sum_i [G(\cdots + \hat{\beta}_k(\mathbf{c}_k + 1)) - G(\cdots + \hat{\beta}_k\mathbf{c}_k)]$.

MROZ: Average Partial Effects Reconcile the Models

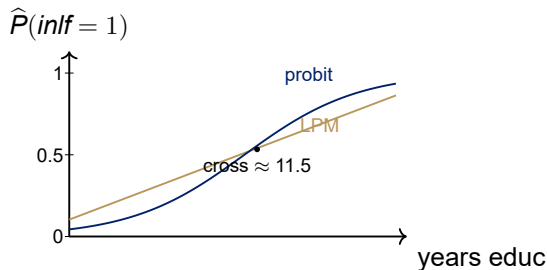
Table 17.2 — APEs for labor force participation

Variable	LPM	Logit	Probit
<i>nwifeinc</i>	−0.0034	−0.0038	−0.0036
<i>educ</i>	0.038	0.039	0.039
<i>exper</i>	0.027	0.025	0.026
<i>age</i>	−0.016	−0.016	−0.016
<i>kidslt6</i>	−0.262	−0.258	−0.261
<i>kidsge6</i>	0.013	0.011	0.011

The punchline. Once converted to APEs, all three models tell nearly the *same* story — another year of schooling raises participation by about 3.9 percentage points. The LPM was a decent approximation *to the average effect* all along.

Rule of thumb: $\hat{\beta}^{\text{logit}} \approx 1.6 \hat{\beta}^{\text{probit}}$; map to LPM by multiplying probit by ≈ 0.4 , logit by ≈ 0.25 .

Diminishing Effects: Probit vs. LPM Across Education



Recreates **Figure 17.2**.

- ▶ Low education: LPM predicts *higher* participation than probit.
- ▶ Curves cross near 11.5 years.
- ▶ High education: probit pulls ahead.
- ▶ LPM's constant slope is the cost of its simplicity.

Answer to Q1. A probit fitted value is $\Phi(\cdot) \in (0, 1)$ by construction — it can *never* exceed 1, unlike the LPM.

Fractional Response Models (§17-2)

Fractional Outcomes: Building the Model

A **fractional response** takes values in $[0, 1]$, possibly including the endpoints: the fraction of employees in a pension plan, the share of votes won, the proportion of students who pass an exam.

The object of interest is the **conditional mean** $\mathbb{E}(y \mid \mathbf{x})$, which must itself lie in $[0, 1]$. A linear model $\mathbb{E}(y \mid \mathbf{x}) = \beta_0 + \mathbf{x}\beta$ can produce means outside $[0, 1]$ — exactly the LPM defect, now for a *mean* rather than a probability.

Fractional response model

Pass the index through a CDF G with $0 < G(z) < 1$:

$$\mathbb{E}(y \mid \mathbf{x}) = G(\beta_0 + \mathbf{x}\beta), \quad y \in [0, 1].$$

Fractional logit: $\mathbb{E}(y \mid \mathbf{x}) = \Lambda(\beta_0 + \mathbf{x}\beta) = \frac{\exp(\beta_0 + \mathbf{x}\beta)}{1 + \exp(\beta_0 + \mathbf{x}\beta)}.$

Fractional probit: $\mathbb{E}(y \mid \mathbf{x}) = \Phi(\beta_0 + \mathbf{x}\beta).$

The *same* functional forms as binary logit/probit — but here $G(\cdot)$ is a conditional **mean**, not a probability. Binary response is the special case $y \in \{0, 1\}$, where the mean *equals* $P(y = 1 \mid \mathbf{x})$.

Fractional Logit: Quasi-Maximum Likelihood Estimation

We cannot use a Bernoulli *likelihood* — y is no longer in $\{0, 1\}$. Yet Papke and Wooldridge ([Papke and Wooldridge, 1996](#)) show that we may maximize the *same* Bernoulli quasi-log-likelihood:

Bernoulli quasi-log-likelihood

$$\ell_i(\beta) = y_i \log[G(\mathbf{x}_i\beta)] + (1 - y_i) \log[1 - G(\mathbf{x}_i\beta)], \quad y_i \in [0, 1].$$

Maximizing $\sum_i \ell_i(\beta)$ gives the **fractional logit** ($G = \Lambda$) or **fractional probit** ($G = \Phi$) estimator.

- ▶ **Quasi-MLE**: the likelihood is knowingly *wrong* (y is not Bernoulli), hence “QMLE” — yet $\hat{\beta}$ is consistent.
- ▶ **Consistency** requires only that the mean $\mathbb{E}(y \mid \mathbf{x}) = G(\beta_0 + \mathbf{x}\beta)$ be correct; the rest of the distribution is left unspecified.
- ▶ **Robust SEs** are mandatory: the Bernoulli variance $G(1 - G)$ is incorrect for $y \notin \{0, 1\}$ (it overstates, so robust SEs are smaller).
- ▶ **Partial effects** as in binary: $\partial \mathbb{E}(y \mid \mathbf{x}) / \partial x_j = g(\beta_0 + \mathbf{x}\beta) \beta_j$, with $g(z) = \Lambda(z)[1 - \Lambda(z)]$ for fractional logit.

Example: 401(k) Participation Rates

Table 17.3 — *prate* participation rate ($n = 4075$)

Variable	Linear OLS	Frac. Logit	Frac. Logit APE
<i>mrte</i>	0.107 (0.006)	1.158 (0.075)	0.147 (0.010)
<i>age</i>	0.0037 (0.0002)	0.035 (0.003)	0.0044 (0.0004)
<i>ltotemp</i>	-0.028 (0.002)	-0.207 (0.014)	-0.026 (0.002)
<i>sole</i>	0.018 (0.006)	0.166 (0.051)	0.021 (0.006)
R^2	0.176	0.195	—

The linear model produces 159 fitted values *above one*; fractional logit cannot. Its match-rate APE (0.147) exceeds the OLS slope (0.107), capturing a *diminishing* effect a linear model needs an ad-hoc $mrte^2$ to mimic.

Nonnegative Outcomes: Poisson Regression (§17-3)

Counts and Nonnegative y : The Exponential Mean

When $y \in \{0, 1, 2, \dots\}$ (arrests, patents) or any $y \geq 0$, a linear model can predict negatives. Use an **exponential conditional mean**:

$$\mathbb{E}(y \mid \mathbf{x}) = \exp(\beta_0 + \mathbf{x}\boldsymbol{\beta}) > 0.$$

Taking logs, $\log \mathbb{E}(y \mid \mathbf{x}) = \beta_0 + \mathbf{x}\boldsymbol{\beta}$, so $100 \beta_j$ is approximately the **percentage change** in $\mathbb{E}(y \mid \mathbf{x})$ for a one-unit rise in x_j — familiar semi-elasticity interpretation, but *valid even when $y = 0$* .

Why not just use $\log(1 + y)$? Because it is *not* scale-invariant in y and does not give clean percentage-change coefficients (Chen and Roth, 2024). The exponential mean sidesteps the “log of zero” problem entirely.

Poisson Regression and the QMLE Escape Hatch

Poisson model

Assume $y \mid \mathbf{x}$ is Poisson with mean $\exp(\mathbf{x}\beta)$:

$$P(y = h \mid \mathbf{x}) = \frac{\exp[-\exp(\mathbf{x}\beta)] [\exp(\mathbf{x}\beta)]^h}{h!}, \quad h = 0, 1, 2, \dots$$

Maximize $\mathcal{L}(\beta) = \sum_i \{y_i \mathbf{x}_i \beta - \exp(\mathbf{x}_i \beta)\}$.

The Poisson assumption forces $\text{Var}(y \mid \mathbf{x}) = \mathbb{E}(y \mid \mathbf{x})$ — often violated (**overdispersion**, $\sigma^2 > 1$).

Poisson QMLE robustness. $\hat{\beta}$ is *consistent and asymptotically normal whenever the mean $\exp(\mathbf{x}\beta)$ is correct* — regardless of the true variance, even if y is not a count. Just inflate SEs by $\hat{\sigma}$, with $\hat{\sigma}^2 = (n - k - 1)^{-1} \sum_i \hat{u}_i^2 / \hat{y}_i$. This is why Poisson beats negative binomial (Santos Silva and Tenreiro, 2006).

Example: Number of Arrests (CRIME1)

Table 17.5 — Dependent variable: *narr86* (arrests in 1986)

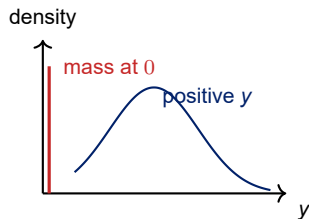
Variable	Linear (OLS)	Exponential (Poisson QMLE)
<i>pcnv</i>	−0.132 (0.040)	−0.402 (0.085)
<i>ptime86</i>	−0.041 (0.009)	−0.099 (0.021)
<i>inc86</i>	−0.0015 (0.0003)	−0.0081 (0.0010)
<i>black</i>	0.327 (0.045)	0.661 (0.074)
Log-likelihood	—	−2,248.76
$\hat{\sigma}$	—	1.232

narr86 is 0 for 1,970 of 2,725 men — a count with a huge zero pileup, ideal for Poisson. A higher conviction probability ($\Delta pcnv = 0.1$) lowers expected arrests by about 4%; a black man's expected arrests are $100[\exp(0.661) - 1] \approx 94\%$ higher, ceteris paribus. Since $\hat{\sigma} = 1.232 > 1$ (overdispersion), inflate SEs by $\approx 23\%$.

Corner Solutions: The Tobit Model (§17-4)

Corner Solutions: A Pileup of Zeros

Hours worked, charitable gifts, insurance: a **nontrivial fraction choose exactly zero**, then a continuous spread of positives. OLS predicts negatives, cannot match the spike at zero, and is heteroskedastic.



Tobit model (Tobin, 1958)

A latent linear model, observed only when positive:

$$y^* = \beta_0 + \mathbf{x}\boldsymbol{\beta} + u, \quad u \mid \mathbf{x} \sim \mathcal{N}(0, \sigma^2), \quad y = \max(0, y^*).$$

Tobit: Likelihood and the Role of σ

Two pieces: a **probability** of the corner and a **density** for positives.

$$P(y = 0 \mid \mathbf{x}) = 1 - \Phi(\mathbf{x}\boldsymbol{\beta}/\sigma), \quad y > 0 : \frac{1}{\sigma}\phi((y - \mathbf{x}\boldsymbol{\beta})/\sigma).$$

Tobit log-likelihood

$$\ell_i(\boldsymbol{\beta}, \sigma) = \mathbf{1}[y_i = 0] \log[1 - \Phi(\mathbf{x}_i\boldsymbol{\beta}/\sigma)] + \mathbf{1}[y_i > 0] \log\left\{\frac{1}{\sigma}\phi((y_i - \mathbf{x}_i\boldsymbol{\beta})/\sigma)\right\}.$$

Estimate $\boldsymbol{\beta}$ and σ jointly by MLE. σ is **not “ancillary”** — it enters the partial effects directly, so it matters for the economics, not just for inference.

The Inverse Mills Ratio: A Truncated-Normal Lemma

Every Tobit (and selection) expectation rests on one fact about the normal.

Truncated-normal lemma

If $z \sim \mathcal{N}(0, 1)$, then for any constant c ,

$$\mathbb{E}[z \mid z > c] = \frac{\phi(c)}{1 - \Phi(c)}, \quad \mathbb{E}[z \mid z > -c] = \frac{\phi(c)}{\Phi(c)} \equiv \lambda(c),$$

where the second form uses $\phi(-c) = \phi(c)$ and $1 - \Phi(-c) = \Phi(c)$.

Inverse Mills ratio $\lambda(c) = \phi(c)/\Phi(c)$ — the “selection hazard”:

- ▶ $\lambda(c) > 0$ for all c , and is **strictly decreasing**.
- ▶ $\lambda(c) \rightarrow 0$ as $c \rightarrow +\infty$; $\lambda(c) \approx -c$ as $c \rightarrow -\infty$, so $c + \lambda(c) > 0$ always.
- ▶ Derivative: $\lambda'(c) = -\lambda(c)[c + \lambda(c)] \in (-1, 0)$.

This one function does the work in *both* the Tobit conditional mean $\mathbb{E}(y \mid y > 0, \mathbf{x})$ and the Heckman selection correction. Watch for it twice.

Tobit Partial Effects via the Inverse Mills Ratio

The β_j measure effects on the *latent* y^* — rarely the goal; we want effects on the **observed** y . Apply the lemma with $z = u/\sigma$, since $y > 0 \Leftrightarrow u/\sigma > -\mathbf{x}\beta/\sigma$:

$$\mathbb{E}(y \mid y > 0, \mathbf{x}) = \mathbf{x}\beta + \sigma \mathbb{E}\left[\frac{u}{\sigma} \mid \frac{u}{\sigma} > -\frac{\mathbf{x}\beta}{\sigma}\right] = \mathbf{x}\beta + \sigma \lambda(\mathbf{x}\beta/\sigma).$$

Combine with $P(y > 0 \mid \mathbf{x}) = \Phi(\mathbf{x}\beta/\sigma)$; the λ term cancels:

$$\mathbb{E}(y \mid \mathbf{x}) = \Phi(\mathbf{x}\beta/\sigma) \mathbb{E}(y \mid y > 0, \mathbf{x}) = \Phi(\mathbf{x}\beta/\sigma) \mathbf{x}\beta + \sigma \phi(\mathbf{x}\beta/\sigma).$$

Hence the partial effect on the **unconditional** mean is remarkably clean:

$$\frac{\partial \mathbb{E}(y \mid \mathbf{x})}{\partial x_j} = \beta_j \Phi(\mathbf{x}\beta/\sigma).$$

The factor $\Phi(\mathbf{x}\beta/\sigma) \in (0, 1)$ scales the latent coefficient down to the observed-mean effect — the APE scale factor for Tobit.

MROZ: Annual Hours Worked — OLS vs. Tobit

Table 17.6/17.7 — Dependent variable: *hours* (325 zeros)

Variable	Coefficients		APE	
	OLS	Tobit	OLS	Tobit
<i>nwifeinc</i>	−3.45 (2.24)	−8.81 (4.46)	−3.45	−5.19
<i>educ</i>	28.76 (13.04)	80.65 (21.58)	28.76	47.47
<i>age</i>	−30.51 (4.24)	−54.41 (7.42)	−30.51	−32.03
<i>kidslt6</i>	−442.1 (57.5)	−894.0 (111.9)	−442.1	−526.3
$\hat{\sigma}$	750.18	1,122.02		

Do not compare raw coefficients. The Tobit *educ* coefficient (80.65) dwarfs OLS (28.76). But the APE scale factor $n^{-1} \sum_i \Phi(\mathbf{x}_i \beta / \hat{\sigma}) \approx 0.589$ gives a Tobit APE of $0.589 \times 80.65 \approx 47.5$ hours — larger than OLS, but far less dramatic than the raw coefficient suggests.

When Is Tobit the Wrong Model?

Tobit ties $P(y > 0 \mid \mathbf{x})$ and $\mathbb{E}(y \mid y > 0, \mathbf{x})$ to the *same* parameters β/σ . A regressor cannot raise the chance of a positive outcome while lowering its size.

Specification check. Run a probit of $w = \mathbf{1}[y > 0]$ on $\mathbf{x}$; its coefficients should be “close” to $\hat{\beta}_j/\hat{\sigma}$ from Tobit. Sign flips or large magnitude gaps signal that Tobit is too restrictive — consider a two-part (hurdle) model instead.

Answer to Q2. In MROZ, “hours” is a **corner solution**, not censoring — women genuinely *choose* zero. Tobit is appropriate, and σ is economically meaningful (not a nuisance).

Censored and Truncated Regression (§17-5)

Censoring vs. Truncation: A Data Problem, Not a Behavior

Censored regression

We **always observe** $\mathbf{x}_i$ for a random draw, but y_i is only recorded up to a threshold c_i :

$$y_i = \beta_0 + \mathbf{x}_i\beta + u_i, \quad w_i = \min(y_i, c_i).$$

Example: top-coded wealth (“> \$500k”), duration data.

Truncated regression

A **sub-population is excluded entirely** — no $\mathbf{x}_i$, no y_i — based on y :

observe $(\mathbf{x}_i, y_i)$ only if $y_i \leq c_i$.

Example: a study that only enrolls families below $1.5\times$ the poverty line.

Key difference. Censoring keeps the unit (we know $y > c$); truncation *drops* it. Censored regression exploits the censored units; truncated regression cannot.

Censored Normal Regression: Likelihood

Under $y_i = \beta_0 + \mathbf{x}_i\boldsymbol{\beta} + u_i$, $u_i \mid \mathbf{x}_i, \mathbf{c}_i \sim \mathcal{N}(0, \sigma^2)$ and $w_i = \min(y_i, \mathbf{c}_i)$:

Density of the observed w_i

$$f(w \mid \mathbf{x}_i, \mathbf{c}_i) = \begin{cases} 1 - \Phi((\mathbf{c}_i - \mathbf{x}_i\boldsymbol{\beta})/\sigma), & w = \mathbf{c}_i \quad (\text{censored}), \\ \frac{1}{\sigma}\phi((w - \mathbf{x}_i\boldsymbol{\beta})/\sigma), & w < \mathbf{c}_i \quad (\text{uncensored}). \end{cases}$$

- ▶ Maximize $\sum_i \log f(\cdot)$ to get $\hat{\boldsymbol{\beta}}, \hat{\sigma}$.
- ▶ **Crucial:** here the β_j are interpreted *just as in a linear regression* — no inverse-Mills gymnastics. The expectations of interest are linear in $\boldsymbol{\beta}$.
- ▶ Naively running OLS on *uncensored* obs only $\Rightarrow$ inconsistent (same algebra as Tobit, different cause).

Example: Duration of Recidivism (RECID)

Table 17.8 — DV $\log(\text{durat})$ — 62% censored

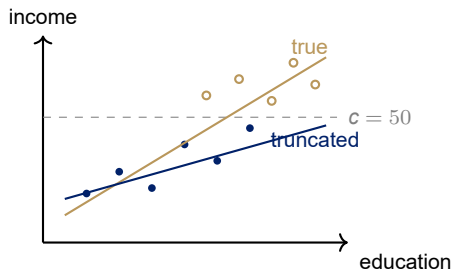
Variable	Coef. (SE)
<i>workprg</i>	−0.063 (0.120)
<i>priors</i>	−0.137 (0.021)
<i>tserved</i>	−0.019 (0.003)
<i>felon</i>	0.444 (0.145)
<i>alcohol</i>	−0.635 (0.144)
<i>married</i>	0.341 (0.140)
$\hat{\sigma}$	1.810

durat = months to re-arrest; 893 of 1,445 never re-arrested $\Rightarrow$ **censored**.

- ▶ One more prior $\Rightarrow \approx 14\%$ shorter duration.
- ▶ *alcohol* abuse sharply shortens duration.
- ▶ *workprg* effect small, insignificant.

Ignoring censoring (OLS treating censored as actual) shrinks estimates toward zero: *priors* goes from -0.137 to -0.059 . Censored regression recovers the correct magnitudes.

Truncation Biases OLS Toward Zero



Recreates **Figure 17.4**. Dropping incomes above \$50k (hollow points) lops off the upper-right tail; the fitted line through the survivors **flattens** — biased toward zero.

The **truncated normal MLE** restores consistency by re-normalizing the density over the observed region (Hausman and Wise, 1977).

Sample Selection Corrections (§17-6)

When Is OLS on a Selected Sample Still OK?

Let $s_i = 1$ if observation i is used, $s_i = 0$ otherwise. We run OLS on the $s_i = 1$ sub-sample.

Exogenous selection — OLS is fine

- ▶ s depends only on $\mathbf{x}$ (or on factors independent of u)
- ▶ Random sub-sampling
- ▶ $\Rightarrow \hat{\beta}$ still consistent / unbiased

Endogenous selection — OLS biased

- ▶ s depends on y (truncation on the outcome)
- ▶ s depends on u (unobservables)
- ▶ $\Rightarrow$ selection bias

Rule. Selecting on the explanatory variables is harmless. Selecting on the dependent variable (or on the error) is poison. The danger case is **incidental truncation**.

Incidental Truncation: The Wage-Offer Problem

We want the **wage offer** equation for *all* women, but we observe a wage only for those who *choose to work*.

Selection model (Heckman, 1979)

$$\begin{aligned}\text{outcome: } & y = \mathbf{x}\beta + u, & \mathbb{E}(u \mid \mathbf{x}, \mathbf{z}) &= 0, \\ \text{selection: } & s = \mathbf{1}[\mathbf{z}\gamma + v \geq 0], & v &\sim \mathcal{N}(0, 1).\end{aligned}$$

We observe y only when $s = 1$. Truncation of y is *incidental* — it depends on another variable (working), not on y itself.

If $\text{Corr}(u, v) = \rho \neq 0$, the working sample is **not** representative: women who work have systematically different u . OLS on workers is then inconsistent for β .

The Selection Bias as an Omitted Variable

Take the expectation of $y = \mathbf{x}\beta + u$ given $(\mathbf{z}, v)$. With $\mathbf{x} \subseteq \mathbf{z}$, (u, v) independent of $\mathbf{z}$, and (u, v) jointly normal so that $\mathbb{E}(u \mid v) = \rho v$:

$$\mathbb{E}(y \mid \mathbf{z}, v) = \mathbf{x}\beta + \mathbb{E}(u \mid v) = \mathbf{x}\beta + \rho v.$$

Now condition on selection, $s = 1 \Leftrightarrow v > -\mathbf{z}\gamma$. By the **truncated-normal lemma**, $\mathbb{E}(v \mid v > -\mathbf{z}\gamma) = \lambda(\mathbf{z}\gamma)$:

$$\mathbb{E}(y \mid \mathbf{z}, s = 1) = \mathbf{x}\beta + \rho \lambda(\mathbf{z}\gamma), \quad \lambda(\mathbf{c}) = \frac{\phi(\mathbf{c})}{\Phi(\mathbf{c})}.$$

OLS of y on $\mathbf{x}$ using workers only **omits** the regressor $\rho \lambda(\mathbf{z}\gamma)$. Since $\lambda(\cdot)$ correlates with $\mathbf{x}$, this is classic **omitted-variable bias** — vanishing only when $\rho = 0$.

The fix writes itself: *construct* the missing regressor $\lambda(\mathbf{z}\gamma)$ and add it back. OLS on the selected sample is then consistent — precisely Heckman's two-step idea.

The Heckit Procedure

Heckman two-step (Heckit)

1. **Selection equation.** Using *all* n observations, estimate a probit of s_i on $\mathbf{z}_i$. Form the estimated inverse Mills ratio

$$\hat{\lambda}_i = \lambda(\mathbf{z}_i \hat{\gamma}) = \phi(\mathbf{z}_i \hat{\gamma}) / \Phi(\mathbf{z}_i \hat{\gamma}).$$

2. **Outcome equation.** Using only the selected sample ($s_i = 1$), run

$$y_i \text{ on } \mathbf{x}_i, \hat{\lambda}_i.$$

- ▶ The t statistic on $\hat{\lambda}_i$ is a **direct test for selection bias** ($H_0 : \rho = 0$).
- ▶ **Exclusion restriction:** $\mathbf{z}$ should contain at least one variable *not* in $\mathbf{x}$ — something that shifts the decision to work but not the wage offer. Without it, multicollinearity inflates the SEs.

MROZ: Wage Offer Equation — OLS vs. Heckit

Table 17.9 — DV $\log(\text{wage})$ — 428 workers

Variable	OLS	Heckit
<i>educ</i>	0.108 (0.014)	0.109 (0.016)
<i>exper</i>	0.042 (0.012)	0.044 (0.016)
<i>exper</i> ²	−0.00081 (0.00039)	−0.00086 (0.00044)
$\hat{\lambda}$	—	0.032 (0.134)
<i>n</i>	428	428

Excluded from *x* but in *z*: *nwifeinc*, *age*, *kidslt6*, *kidsge6* (drive participation, assumed not the wage offer).

Answer to Q3. Here $t_{\hat{\lambda}} = 0.032/0.134 \approx 0.24$ — we *fail to reject* $\rho = 0$. No evidence of selection bias; OLS and Heckit returns to education differ by one-tenth of a point. Testing first saved an unnecessary correction.

Synthesis

Decision Guide: Which Model for Which y ?

Nature of y	Model	Key interpretation tool
Binary $\{0, 1\}$	Logit / Probit (MLE)	APE / PEA
Fractional $[0, 1]$	Fractional logit (QMLE)	APE, robust SE
Count / nonneg. $y \geq 0$	Poisson (QMLE)	$100\beta_j \approx \% \Delta \mathbb{E}(y)$
Corner solution $y \geq 0$	Tobit (MLE)	$\text{APE} = \Phi(\mathbf{x}\beta/\sigma)\beta_j$
Top-coded / duration	Censored normal reg.	β_j as in linear model
Sub-population excluded	Truncated normal reg.	β_j as in linear model
Selection on outcome	Heckit (two-step)	inverse Mills + t -test on ρ

First question to ask: is y *genuinely* limited in the population (behavior $\rightarrow$ logit/probit/Poisson/Tobit), or is it fine but *badly observed* (data $\rightarrow$ censored/truncated/Heckit)?

Recurring Themes Across All Models

1. Coefficients $\neq$ effects. In every nonlinear model the raw β_j must be transformed (APE, percentage change, Mills adjustment) before it answers an economic question.

2. MLE and QMLE do the heavy lifting. Logit, probit, Tobit, censored, and truncated models are all MLE; fractional and Poisson are *quasi*-MLE — consistent for the mean even when the full distribution is wrong, provided SEs are made robust.

3. Selection is omitted-variable bias. The inverse Mills ratio is the missing regressor; estimate it, add it back, and test whether it mattered.

Summary

Binary & fractional (§17-1, §17-2). Logit/probit squash the index into $(0, 1)$; estimate by MLE/QMLE; summarize with APEs. The LPM approximates the average effect well.

Nonnegative (§17-3). Exponential mean + Poisson QMLE handles counts and zeros, gives percentage-change coefficients, and is robust to overdispersion.

Corner solutions (§17-4). Tobit models the zero pileup; partial effects scale the latent coefficients by $\Phi(\mathbf{x}\beta/\sigma)$.

Data problems (§17-5, §17-6). Censored/truncated regression fix observability defects; Heckit corrects incidental-truncation selection bias and provides a built-in test for it.

Bridge: From LDVs to Causal Inference

Lecture 17 (today)

Modeling *limited* outcomes:

- ▶ Respect the range of y
- ▶ Recover partial effects, not just coefficients
- ▶ Correct for non-random observation

Next: Causal Inference (Ch. 19)

Designs for credible effects:

- ▶ Treatment effects with binary/limited outcomes
- ▶ RD, DiD extensions, matching
- ▶ Potential outcomes meet nonlinear models

The logit/probit and Tobit machinery returns in causal settings: average treatment effects with nonlinear models are just APEs of a treatment dummy. The threads from today run straight into the next lecture.

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